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The intrinsically disordered C-terminal region of *Arabidopsis thaliana* TCP8 transcription factor acts both as a transactivation and self-assembly domain†‡

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TCPs are plant specific transcription factors with non-canonical basic helix–loop–helix domains. While *Arabidopsis thaliana* has 24 TCPs involved in cell proliferation and differentiation, their mode of action has not been fully elucidated. Using bioinformatic tools, we demonstrate that TCP transcription factors belong to the intrinsically disordered proteins (IDP) family and that disorder is higher in class I TCPs than in class II TCPs. In particular, using bioinformatic and biochemical approaches, we have characterized TCP8, a class I TCP. TCP8 exhibits three intrinsically disordered regions (IDR) made of more than 50 consecutive residues, in which phosphorylable Ser residues are mainly clustered. Phosphorylation of Ser-211 that belongs to the central IDR was confirmed by mass spectrometry. Yeast two-hybrid assays also showed that the C-terminal IDR corresponds to a transactivation domain. Moreover, biochemical experiments demonstrated that TCP8 tends to oligomerize in dimers, trimers and higher-order multimers. Bimolecular fluorescence complementation (BiFC) experiments carried out on a truncated form of TCP8 lacking the C-terminal IDR indicated that it is effectively required for the pronounced self-assembly of TCP8. These data were reinforced by the prediction of a coiled coil domain in this IDR. The C-terminal IDR acts thus as an oligomerization domain and also a transactivation domain. Moreover, many Molecular Recognition Features (MoRFs) were predicted, indicating that TCP8 could interact with several partners to fulfill a fine regulation of transcription in response to various stimuli.

Introduction

Until the early 1990s, it was almost exclusively accepted that the function of a protein was linked to its three-dimensional structure. This concept is undergoing re-evaluation with the description of intrinsic disorder. Proteins can be wholly disordered (intrinsically disordered proteins, IDPs) or they can contain intrinsically disordered regions (IDR). They are then characterized by low compactness, absence of globularity, low secondary structure content and high flexibility.[1] Such proteins are present across the three kingdoms of life but their mean frequency is higher in Eukaryotes.[2] Indeed, about 25 to 30% of eukaryotic proteins are mostly disordered and more than 50% have long regions of disorder, while bacteria and archae are predicted to have 4–45% and 2–51% of their proteins with long disordered regions, respectively.[3–6] The frequency reaches 70 and 90% with eukaryotic signaling proteins and transcription factors, respectively.[7,8] The predominant cellular functions of long disordered regions appear to involve molecular recognition and in particular, binding of DNA to facilitate processes such as transcription, transposition or replication. Signaling, cell cycle, development and endocytosis are also processes where IDR are especially common.[6,9–12] IDP and IDR can recognize and bind several partners at various binding sites,[13–16] as exemplified by hub proteins.[17] They may bind to different partners *via* short regions, named Molecular Recognition Features (MoRFs) that undergo disorder-to-order transitions.[18]

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However, IDPs can also keep disordered regions in the bound state with various degrees of dynamics, forming so-called "fuzzy complexes".[19,20]

Numerous IDP have been characterized in the animal kingdom and particularly in human diseases,[21–23] but there have only been a few reports from the plant kingdom.[24–28] Protein intrinsic disorder is thus an emerging topic in plant science. For example, a chloroplast IDP named CP12 plays key roles as a hub protein in plant metabolism.[28–30] The C-terminal domain of cryptochromes is also a crucial IDR in plant signaling.[31] The chaperones ERD10 and ERD14, that belong to the late embryogenesis abundant (LEA) class, have been shown to be fully disordered[24,32] and LEA proteins are currently the plant protein family with the largest number of known IDPs.[26] *In silico* analysis of 84 plant specific remorin proteins has shown a clear tendency to disorder and it was experimentally confirmed for one of them, AtREM1.3.[33,34] Recently, the N-terminal domain of DELLA proteins that belong to the GRAS proteins family, was characterized to be disordered and further bioinformatic studies have shown that it is true for the N-domains of all GRAS proteins.[25,35] RCD1 (Radical-induced cell death 1) interacts with several transcription factors (TF) and all these interactions imply IDRs in the TFs.[27] IDRs are effectively prevalent in TFs and allow the recognition of multiple protein partners thus enabling a fine regulation of the transcriptional machinery.[8,12,36,37] The importance of such IDR in TFs has been extensively described for the bZIP TF and NAC TF superfamily.[37,38]

The TCP family of plant specific TFs, named after its 3 founding members (*TEOSINTE BRANCHED1*, *CYCLOIDEA* (CYC),

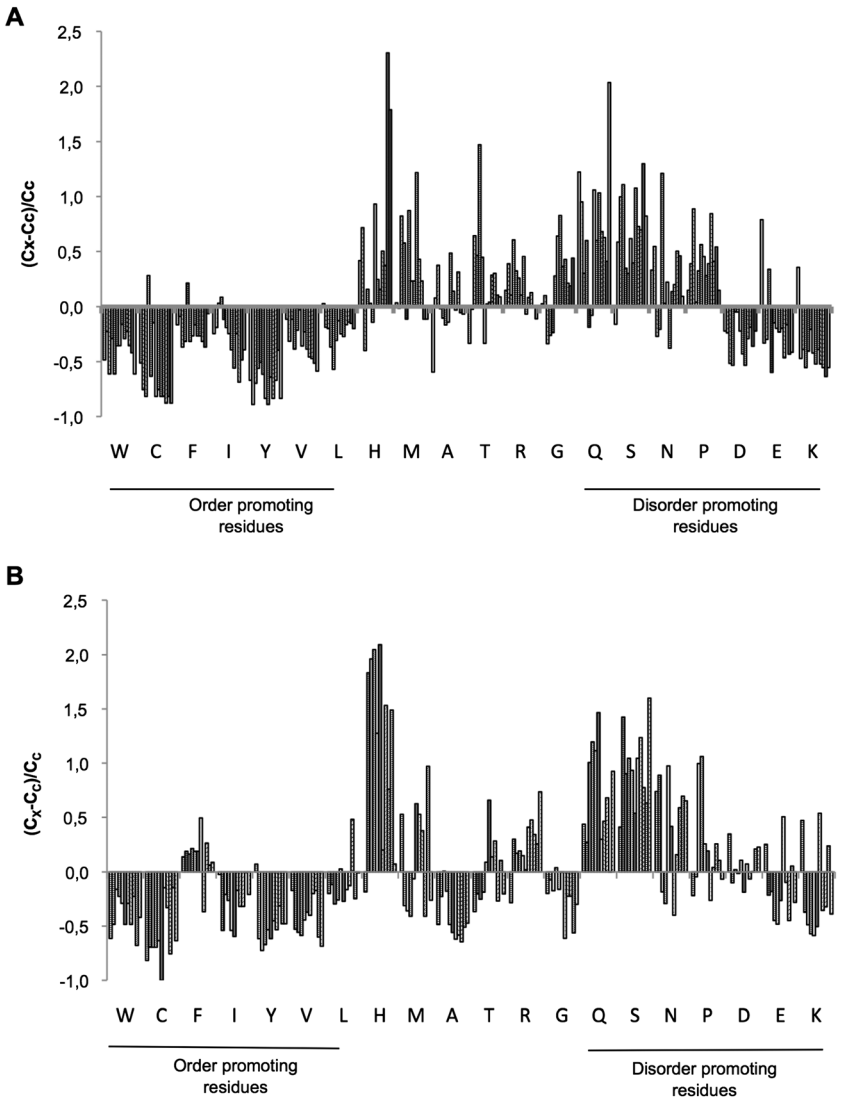

Fig. 1 Amino acid composition profiles of TCP proteins. For a given amino acid, the bars represent the fractional difference in composition between class I TCPs (A) or class II TCPs (B) and the average values in the Swiss-Prot database. The fractional difference is calculated as $(c_x - c_c)/c_c$, where c_x is the composition of a given amino acid in TCPs and c_c is the corresponding average composition in the Swiss-Prot database. For a given amino acid, each bar corresponds to a TCP protein. Class I and class II contained 13 and 11 TCPs, respectively. The amino acids are ordered according to the Vihinen flexibility scale.[96] Negative values indicate residues that are depleted meanwhile positive values indicate residues that are enriched.

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Proliferating Cell Nuclear Factor (PCF)),[39–41] is emerging in plant science. In *Arabidopsis thaliana*, 24 genes encoding TCPs have been identified.[42] TCPs contain a non-canonical basic helix–loop–helix motif in the N-terminal half of the protein known as the TCP domain and can be divided into two classes based on the presence or not of 4 amino acids in the basic domain.[43] TCPs are generally involved in development of vegetative and reproductive organs. Class II members regulate growth in different organs, mainly through the inhibition of cell proliferation or enlargement.[44–46] To date, involvement of class I TCPs has been described in pollen development, germination, organ shape and growth, and inflorescence and flower development.[47–51] These multiple effects of TCPs are probably the consequence of their interaction with multiple partners in several regulatory complexes.[50,52–57]

If TCPs play a crucial role in plant development, their biochemical characterization is still in its infancy. Therefore, the aim of this report was to increase the knowledge about the TCP family, by analyzing their sequences using a range of bioinformatic tools. Biochemical studies have also been performed for the first time on one member, TCP8. This report shows that TCP8 is highly disordered, phosphorylated and has a high propensity to oligomerize. This work also demonstrates that the C-terminal IDR is a transactivation domain and is involved in TCP8 self-assembly.

Results

TCPs belong to the intrinsically disordered proteins family

The amino acid compositional bias within proteins is a signature of probable disordered proteins.[58] The amino acid composition of all the TCPs was thus systematically compared with the average values recorded for amino acid composition of proteins from the SwissProt database. Both class I and class II TCPs were poor in order-promoting residues Trp, Cys, Ile, Tyr and Val (Fig. 1). In contrast, they were enriched in the disorder-promoting residues Gln, Ser, Asn and Pro. More variability was observed with Asp, Glu and Lys. While class I TCPs were mainly depleted in these residues, no consensus appeared with class II TCPs. All these data indicate that the amino acid composition of TCPs is typical of IDPs. Moreover, all the members of the TCP family were analyzed using PONDR VL-XT predictor (http://www.pondr.com). PONDR predicted a degree of disorder ranging from 31.5 to 73.3% (Table 1). The highest percentages of disorder were observed for TCP8, TCP9, TCP22 and TCP23, belonging to class I TCPs that on average are more disordered than class II TCPs. The lowest percentage of disorder was predicted for TCP16. The length of the disordered regions varied also among TCP members from 27 to 98 residues. The highest lengths were observed for TCP9 (98 residues) and TCP12 (90 residues). The distribution and the length of IDRs over the sequence were specific of each TCP (Fig. S1, ESI†).

The Coils algorithm was used to check whether coiled coils (CC) are present in TCPs since CC are often linked to intrinsic disorder[59] and Gln/Asn- and Gln-rich domains are disordered structures prone to form coiled coils.[60] Three TCPs belonging to class I (*i.e.* TCP6, TCP8, and TCP11) and one class II TCP (TCP5) were predicted with a high probability (>0.8) to contain a CC domain (Table 2).

Table 1 Overall disorder in TCP transcription factors family. Class I TCPs are highlighted in italic font. The most disordered TCPs are indicated in bold

	Overall percent of disorder	Number of disordered regions	Number of residues in the longest disordered region
TCP1	47.63	6	59
TCP2	41.10	10	58
TCP3	43.73	5	71
TCP4	42.14	8	66
TCP5	35.56	7	41
TCP6	*51.03*	*7*	*39*
TCP7	*51.20*	*4*	*46*
TCP8	*63.34*	*9*	*73*
TCP9	*73.31*	*7*	*98*
TCP10	40.28	9	40
TCP11	*51.06*	*5*	*27*
TCP12	50.84	5	90
TCP13	56.06	6	85
TCP14	*46.22*	*12*	*48*
TCP15	*59.69*	*8*	*51*
TCP16	*31.52*	*5*	*33*
TCP17	36.36	6	34
TCP18	37.07	4	80
TCP19	*59.39*	*7*	*68*
TCP20	*57.06*	*9*	*37*
TCP21	*51.88*	*6*	*31*
TCP22	*62.93*	*12*	*83*
TCP23	*65.40*	*7*	*75*
TCP24	45.06	7	40

Disorder prediction of TCP8, a class I TCP

All these data prompted us to particularly study TCP8 since it was one of the most highly disordered TCPs and it contained a putative CC domain. MeDor, a metaserver that predicts protein disorder thanks to multiple methods (IUPred, FoldIndex and DisEMBL, for example) and provides hydrophobic cluster analysis (HCA) plots (http://vazymolo.org), as well as PONDR predictor were used. Three main disordered regions were predicted whatever the software used (Fig. 2A–B). An IDR of 61 residues (res 19–79; 88% of disorder), at the N-terminal part, overlapped the basic DNA binding region present in the TCP domain. In the middle of the sequence, a segment of 60 residues (res 190–249) was predicted to be fully disordered. Finally, a region of 73 residues (res 329–401), at the C-terminal end of TCP8, was predicted to have an overall percent of disorder of 98.63%. The Coils algorithm predicted, with a high probability (~1), a CC domain in this third disordered area (Fig. 2A). An amino acid compositional bias was also observed in each of these domains. While the N-terminal disordered part is enriched in Thr and Arg, the middle IDR is enriched in His, Gln and Asn and the C terminal part is enriched in Ser, Asn and Gln (Fig. 2A).

MoRFpred (http://biomine.ece.ualberta.ca/MORFpred/) was used to identify MoRFs, implicated in protein–protein interactions. ANCHOR predictor (http://anchor.enzim.hu/), which finds disordered binding regions that become ordered when bound to a globular partner, was also used. Several binding regions and/or MoRFs were predicted in each disordered

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Table 2 Coiled coils (CC) prediction in TCPs using the Coils algorithm (http://www.ch.embnet.org)[a]

$0.8 < p < 1$	$0.6 < p < 0.8$	$0.4 < p < 0.6$	$p < 0.4$
TCP5, TCP6, TCP8, TCP11	TCP10, TCP12	TCP4, TCP13, TCP24	TCP1, TCP2, TCP3, TCP7, TCP9, TCP14, TCP15, TCP17, TCP18, TCP19, TCP20, TCP21, TCP22, TCP23

[a] p corresponds to the probability for a CC domain.

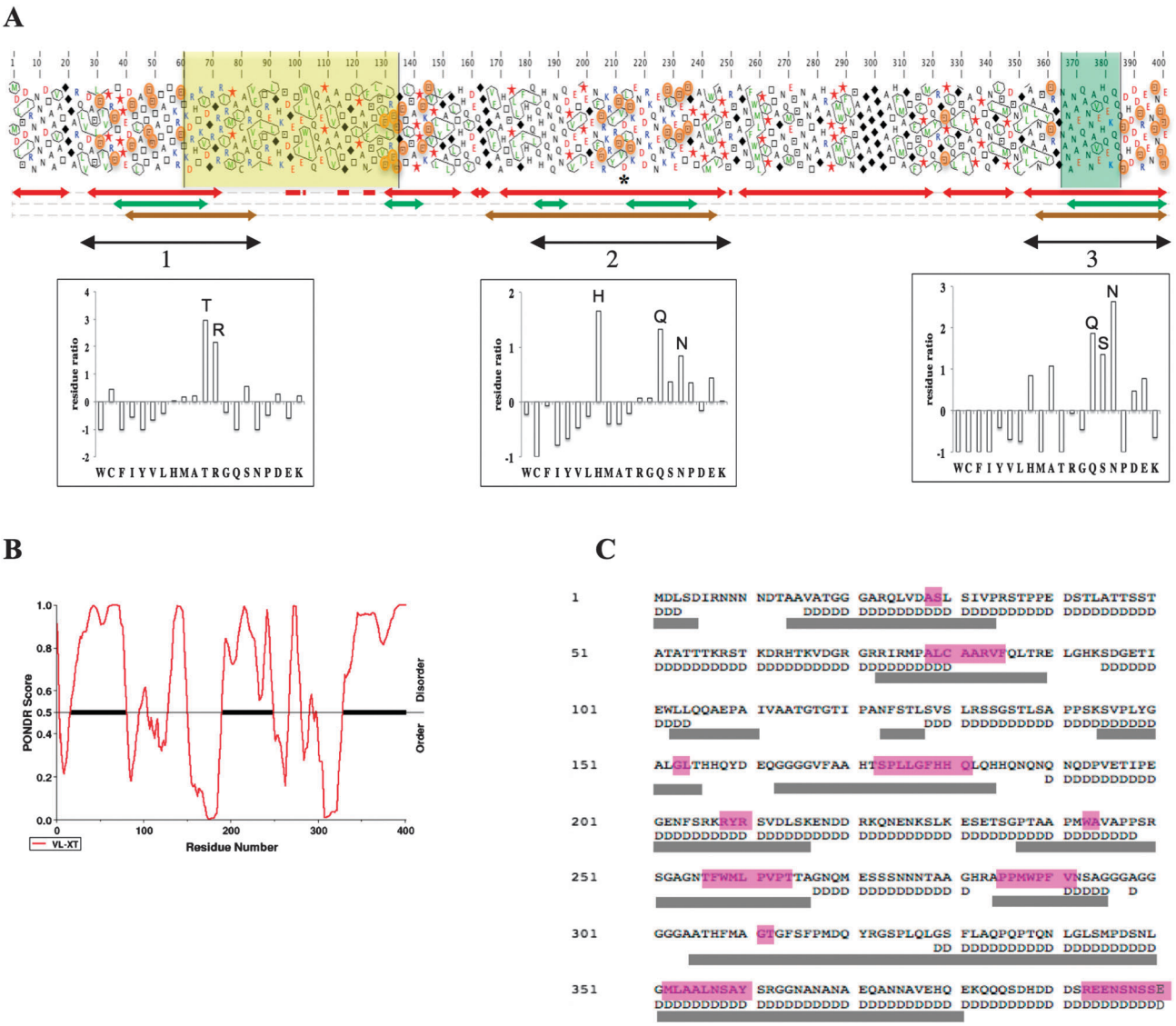

Fig. 2 Disorder prediction for TCP8. (A) Hydrophobic cluster analysis (HCA). HCA plot and disordered regions were generated by the MeDor metaserver, available at http://www.vazymolo.org. The TCP domain is highlighted in yellow. Bidirectional arrows represent the predicted regions of disorder using IUPred (red), DisEMBL REM465 (green) and FoldIndex (brown). Black bidirectional arrows indicate disordered regions predicted by the three predictors. The predicted phosphorylated Ser residues are highlighted in orange. Phosphorylation prediction was performed with DEPP predictor available at http://www.pondr.com. The asterisk indicates the phosphorylated Ser-211 identified by mass spectrometry and the coiled coil domain is highlighted in green. The amino acid composition profiles of each IDR are reported under the bidirectional arrows. (B) PONDR prediction of the order/disorder in TCP8. Disorder-prediction scores were plotted against each amino acid residue. The residues that show values above 0.5 are considered as disordered. A thicker line on the threshold line indicates a region of 40 or more consecutive disordered residues. (C) Molecular recognition motifs (MoRFs) that were predicted with (http://biomine.ece.ualberta.ca/MORFpred/) are highlighted in magenta and protein binding regions predicted with ANCHOR (http://anchor.enzim.hu/) are underlined with grey boxes. D stands for a disordered amino acid predicted by PONDR.

fragment of TCP8 and interestingly, the disordered C-terminal part was predicted to contain two MoRFs that flanked the CC domain (Fig. 2C).

Protein phosphorylation occurs with much higher frequency in disordered regions than in ordered ones,[61] therefore, the composition in Ser, Thr and Tyr residues of TCP8 was analyzed.

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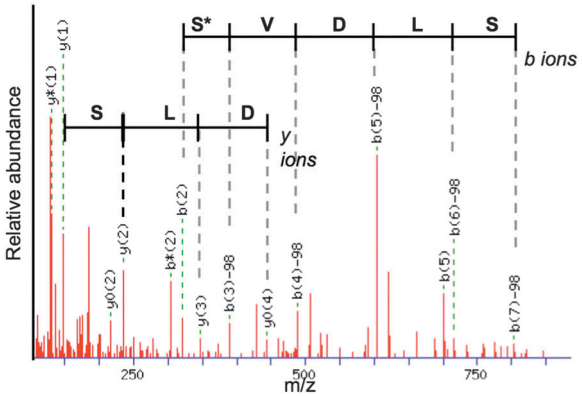

Fig. 3 Mass spectrometry identification of YRSVDLSK phosphorylated peptide in TCP8. Recombinant TCP8, expressed in Hi5 cells, was separated on SDS-PAGE and in-gel digested with trypsin. Resulting peptides were extracted and analyzed by LC-MS/MS analysis on a nanoLC system coupled to a LTQ-Velos Orbitrap mass spectrometer. An asterisk identifies phosphoSer in the *b* ions series. The value −98 indicates the loss of $H_2PO_4^-$ from the phosphorylated peptide.

Tyr residues were twice less abundant than in proteins from the SwissProt database. In contrast, the percentage of Ser residues was 10.7% in TCP8 while a frequency of 6.3% was given for SwissProt proteins. Thr residues were also over-represented, but to a lesser extent, with a percentage of 8% compared to 6.15% in the SwissProt database. 61% of the Ser residues were putative targets for phosphorylation using DEPP Predictor (http://www.pondr.com) and they were mainly clustered in the disordered domains as shown on the HCA plot (Fig. 2A). One of the *in silico* phosphorylation predictions of TCP8 was confirmed using nano liquid chromatography-tandem mass spectrometry (LC-MS/MS). These data showed that at least Ser at position 211 on recombinant TCP8, expressed in Hi5 insect cells, was phosphorylated (Fig. 3). This residue is located in the middle of the second disordered domain and next to a MoRF.

Oligomerization of TCP8

Altogether, *in silico* data showed that TCP8 is a putative IDP and the presence of a CC domain as well as the high frequency of Asn and Gln residues suggested a propensity of TCP8 to oligomerize.[60,62] To confirm these predictions, biochemical experiments were carried out. First, full-length TCP8 cDNA with an N-terminal His-tag sequence was cloned and expressed in *E. coli*. TCP8 was then purified to homogeneity on a Ni-NTA column followed by a DEAE Trisacryl. Recombinant soluble TCP8 protein migrated as a major band at 57 kDa in SDS-PAGE whereas the theoretical calculated molecular mass is 47 kDa (Fig. 4A). MALDI-TOF MS analysis confirmed that this band corresponded to *Arabidopsis* TCP8 (MASCOT score 85). Additional bands of lower molecular mass were revealed by anti-TCP8 antibodies indicating partial degradation (Fig. 4A). The same pattern of migration was obtained with recombinant TCP8 expressed in insect cells (data not shown).

The hydrodynamic properties of TCP8 were determined by gel filtration using a Superdex 200 10/300 GL. The apparent molecular weight (M_r^{App}) and Stokes radius (R_s^{TCP8}) of TCP8 are reported in Table 3. While the M_r^{App} of the monomer was equal to 58.3 kDa and similar to that estimated by SDS-PAGE, the Stokes radius, R_s^{TCP8} was equal to 32.6 Å. Depending on their hydrodynamic characteristics (M_r^{App}, R_s^{TCP8}), IDPs can be divided into three subclasses: molten globule (MG)-like, premolten globule (PMG)-like and random coil-like.[63,64] The apparent Stokes radius of monomeric TCP8 was then compared to the theoretical Stokes radius expected for each subclass (R_s^{MG}, R_s^{PMG} and R_s^{coil}). The TCP8 monomer exhibits a Stokes radius similar to the expected value of molten globule-like IDP.

The oligomeric state of TCP8 was determined using chemical cross-linking with glutaraldehyde. Purified TCP8 was incubated with 0.05% glutaraldehyde, aliquots were taken as a function of time and analyzed on SDS-PAGE and probed with anti-TCP8 antibodies. The control without glutaraldehyde

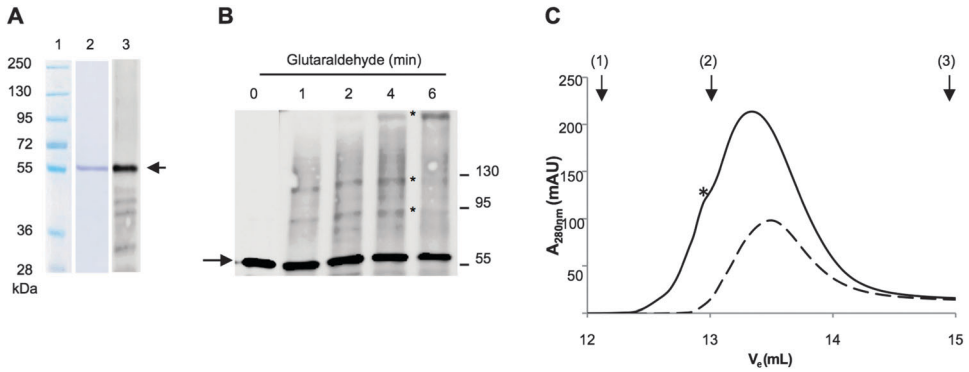

Fig. 4 Biochemical characterization of TCP8. (A) SDS-PAGE of recombinant His-tagged TCP8 expressed in *E. coli*. Proteins were separated on 8% SDS-PAGE gel and stained with Coomassie Brilliant Blue R250 (lane 2) or probed with anti-TCP8 antibodies (lane 3). Lane 1, molecular mass markers. Arrow indicates recombinant His-tagged TCP8. (B) Glutaraldehyde cross-linking of TCP8. Recombinant TCP8 was purified to homogeneity, incubated with 0.05% glutaraldehyde. Proteins (0.5 μg) were withdrawn at a function of time, loaded on SDS-PAGE and subsequently probed with anti-TCP8 antibodies. Arrow stands for TCP8 monomer and asterisks indicate dimer, trimer and higher-order oligomers. (C) Gel filtration elution profile of TCP8 at 4 mg ml^{-1} (solid line) and 1 mg ml^{-1} (dashed line). Asterisk indicates the shoulder corresponding to a trimer. Arrows indicate the elution position of molecular mass markers: 1, β-amylase (200 kDa); 2, alcohol dehydrogenase (150 kDa) and 3, ovalbumin (43 kDa).

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Table 3 Determined apparent molecular weight and Stokes radius of TCP8[a]

	M_r^{Theo}/ kDa	M_r^{App}/ kDa	R_s^{TCP8}/ Å	R_s^{MG}/ Å	R_s^{PMG}/ Å	R_s^{coil}/ Å	R_s^{NF}/ Å	R_s^{TCP8}/ R_s^{MG}	R_s^{TCP8}/ R_s^{NF}
Monomer	47.4	58.3	32.6	32.3	44.2	56.8	29.6	1.01	1.1
Dimer	94.8	97.4	39	40.7	58.5	79.9	38.3	0.96	1.01

[a] A protein can exist in NF, MG, PMG and native coil states and is characterized by the following relationships of its R_s and M_r^{Theo}: natively folded state ($\log R_s^{NF} = 0.369\log(M_r^{Theo}) - 0.254$); molten globule state ($\log R_s^{MG} = 0.334\log(M_r^{Theo}) - 0.053$); native premolten globule state ($\log R_s^{PMG} = 0.403\log(M_r^{Theo}) - 0.239$) and native coil ($\log R_s^{coil} = 0.493\log(M_r^{App}) - 0.551$).[1,97] M_r^{App} and R_s^{TCP8} are apparent molecular weight and Stokes radius determined using the migration rate values obtained from gel filtration.

produced a major band (57 kDa) corresponding to monomeric TCP8 (Fig. 4B). Cross-linked species with apparent molecular masses of 90 and 125 kDa appeared, corresponding to dimeric and trimeric forms of TCP8. Glutaraldehyde cross-linking is known to affect the migration of disordered proteins on SDS-PAGE by decreasing the hydrodynamic radius and increasing SDS binding. This could explain the small difference between the experimentally observed (90 and 125 kDa) and the theoretical (94 and 141 kDa) apparent molecular masses for dimeric and trimeric states. Additionally, high molecular mass bands that barely migrated inside the gel, were observed after increasing incubation time, suggesting that TCP8 was able to form higher-order oligomers.

Oligomerization of purified TCP8 was further analyzed by size exclusion chromatography using a Superdex 200 10/300 GL. One broad peak corresponding to an apparent molecular mass of ~100 kDa was obtained (Fig. 4C). This molecular mass was consistent with a dimer of TCP8 (calculated mass: 95 kDa). The hydrodynamic properties of the dimer were also determined (Table 3). The R_s^{TCP8} was similar to the theoretical R_s^{NF} value indicating that the TCP8 dimer was in a native folded state. The elution profile was shifted towards higher apparent molecular masses, and a shoulder, corresponding to an apparent molecular mass of ~140 kDa appeared, when a higher amount of TCP8 was applied on the column. This value was roughly three times the molecular mass of the TCP8 monomer, suggesting that this shoulder corresponded to a trimer. The broadness of the peaks indicated polydispersity of TCP8, which was effectively confirmed by dynamic light scattering (DLS) since a polydispersity of 39% was determined. One minor peak was also observed just after the dead volume of the column indicating the presence of aggregates (data not shown).

Bimolecular fluorescence complementation (BiFC) experiments were carried out to reinforce the previous data. In BiFC, Yellow Fluorescent Protein (YFP) was split into two non-fluorescent fragments, (YFPN and YFPC). The putative interacting protein partners were fused to these fragments which, when expressed together, resulted in reconstitution of fluorescent YFP. *Nicotiana benthamiana* leaves were infiltrated with a mixture of *Agrobacterium tumefaciens* transformed with BiFC plasmids carrying the full-length cDNA of *TCP8* in frame with YFPN or YFPC.[65] Reconstitution of YFP fluorescence was examined 3 d after transient coexpression of the protein pairs. Yellow dots or spheres of different sizes were observed throughout the nucleoplasm (Fig. 5A). Leaf cells infiltrated with plasmids with TCP8 at alternative positions in the fusion protein relative to YFP showed the same pattern indicating that the presence of YFP at the N- or C-terminal part of the fusion proteins does not affect the interaction between TCP8 monomers (Fig. S3, ESI†). To check whether this aggregation was due to the 35S promoter and an overexpression of TCP8 monomers, *Nicotiana benthamiana* leaves were infiltrated with a mixture of *Agrobacterium tumefaciens* transformed with BiFC plasmids carrying *TCP8* in frame with *YFPN* (*YFPN-TCP8*) and *TCP15*, belonging to class I TCPs, in frame with *YFPC* (*TCP15-YFPC*), both under the control of the 35S promoter. Fluorescence indicated that TCP8 and TCP15 formed heterodimers but after 3 d, fluorescence was clearly scattered and no aggregation was observed (Fig. 5B). Western blot performed with proteins extracted from infiltrated leaves showed that the amounts of TCP8 and TCP15 monomers were similar. The same pattern of fluorescence was observed whatever the fusion proteins used for TCP8 and TCP15, indicating that aggregation is specific of TCP8 self-oligomerization.

Proteins from *N. benthamiana* leaves over-expressing the full-length cDNA of *TCP8* in frame with *YFPN* or *YFPC*, and *HA* or *c-myc* tags, were extracted under native conditions and analyzed by SDS-PAGE and western blot using anti-TCP8 and -HA antibodies. Both antibodies recognized TCP8 monomers and protein aggregates at the top of the separating gel (Fig. 5C, left panel). Protein extraction performed with 4% SDS containing buffer showed that TCP8 aggregates were mainly resistant to SDS when the C-terminal part of TCP8 was free (Fig. 5C, middle panel). Some dissociation could be observed indicating that SDS extraction weakened TCP8 aggregates. In contrast, when the C-terminus of each TCP8 monomer was involved in the fusion with YFP, the aggregates observed *in vivo* were totally disrupted by SDS extraction (Fig. S4, ESI†). Protein extraction performed with urea and thiourea led only to monomers indicating that TCP8–TCP8 dimers were not covalently bound to form aggregates (Fig. 5C, right panel). The absence of aggregation between TCP8 and TCP15 was confirmed by western blot using anti-TCP8 and -HA antibodies (Fig. 5D).

All these experimental data clearly show that TCP8 exhibits typical IDP properties and pronounced self-association propensity.

The C-terminal intrinsically disordered region of TCP8

BiFC experiments showed that the C-terminal part of TCP8 was involved in the resistance of aggregates to SDS extraction (Fig. S4, ESI†). Moreover, a coiled coil domain and a high frequency of Asn and Gln, two aggregation-prone residues, were found in this C-terminal domain. To check whether this C-terminal disordered region acted as an oligomerization domain, we performed BiFC experiments with a deleted version of TCP8 (δC/TCP8). The last 41 residues were removed in order to disrupt the disordered C-terminal part of TCP8 and *δC/TCP8* was cloned in frame with *YFPC* or *YFPN*. Overviews of transformed *N. benthamiana* epidermal cells showed that

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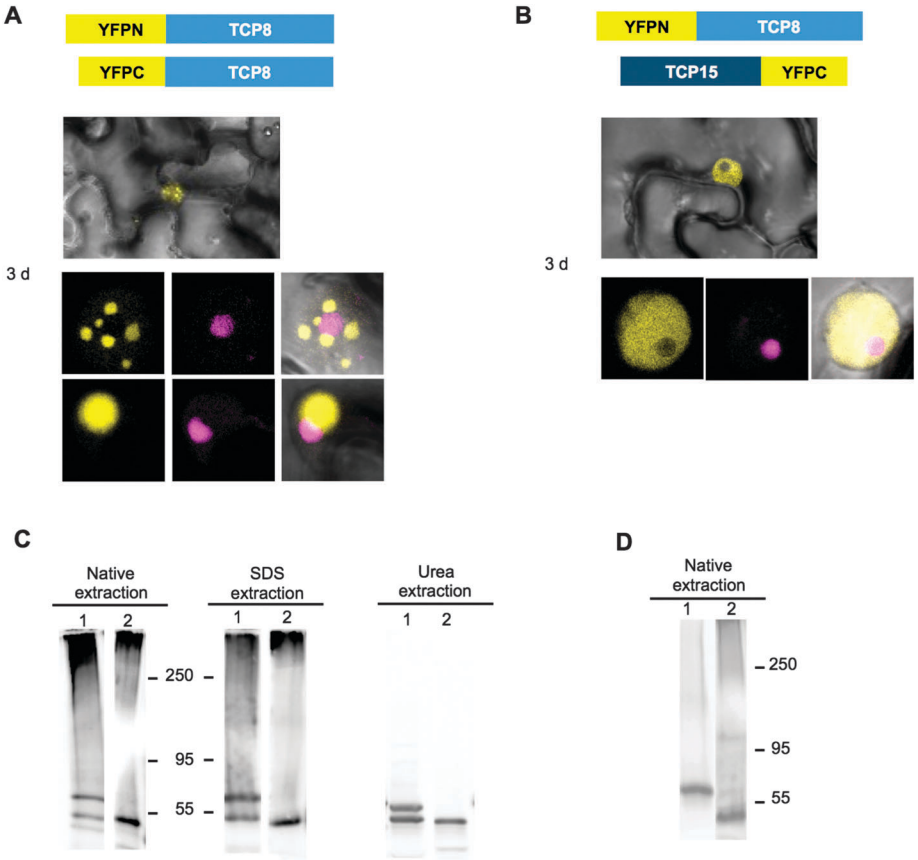

Fig. 5 Aggregation of TCP8. *N. benthamiana* leaves were infiltrated with a mix of *Agrobacterium tumefaciens* transformed with BiFC plasmids carrying the full-length cDNA of *TCP8* in frame with *YFPN* and *YFPC* (A) or with BiFC plasmids carrying the full-length cDNA of *TCP8* in frame with *YFPN* and the full-length cDNA of *TCP15* in frame with *YFPC* (B). For each experiment, *N. benthamiana* leaves were also co-infiltrated with *Agrobacterium tumefaciens* transformed with the pROK2 plasmid carrying *AtFib2-mRFP*. Overviews of YFP fluorescence 3 days after infiltration (top rectangular images), YFP fluorescence (left panels), AtFib2-mRFP fluorescence (middle panels) and merged images (right panels). Controls for the BiFC experiments are shown in Fig. S2 (ESI†). Views are representative of 3 biologically independent experiments. (C) Proteins from infiltrated leaves with YFPC:TCP8 and YFPN:TCP8 constructs were extracted with a native buffer (left panel), with a 4% SDS containing buffer (middle panel) or with a urea and thiourea containing buffer (right panel) and separated by SDS-PAGE and subsequently probed with anti-TCP8 (lanes 1) and anti-HA (lanes 2) antibodies. Anti-TCP8 antibodies recognized TCP8 fused both to C-terminal and N-terminal parts of YFP and anti-HA antibodies recognized the epitope HA only fused to the C-terminal part of YFP. No protein was detected in non-infiltrated leaves (data not shown). Data shown are representative of 2 biologically independent experiments. (D) Proteins from infiltrated leaves with YFPN:TCP8 and TCP15:YFPC constructs were extracted with a native buffer and separated by SDS-PAGE and subsequently probed with anti-TCP8 (lanes 1) and anti-HA (lanes 2) antibodies. Anti-TCP8 antibodies recognized YFPN:TCP8 and anti-HA antibodies recognized the epitope HA fused to TCP15:YFPC. Data shown are representative of 2 biologically independent experiments.

homodimerization of δC/TCP8 still occurred (Fig. 6). Therefore, the C-terminal domain is not essential to TCP8 dimerization. After 3 d of infiltration, while TCP8–TCP8 dimers formed large spheres, the homodimers of δC/TCP8 presented mainly a diffuse nuclear pattern despite the use of a 35S promoter and to a lesser extent, little nuclear speckles (Figs. 6A–B). Proteins from *N. benthamiana* leaves over-expressing the full-length or the truncated cDNA of *TCP8* fused with *YFPN* or *YFPC* were extracted under native conditions and analyzed by SDS-PAGE and western blot. When leaves expressed δC/TCP8, anti-TCP8 antibodies recognized monomers and dimers but no aggregates were observed (Fig. 6C). Therefore, the C-terminal domain of TCP8 is not essential to dimerization but contributes to its self-assembly in high-order oligomers.

A 98.63% value of disorder was predicted for this IDR and the C-terminal part of this region was particularly enriched in acidic residues (Asp and Glu). These data indicated that this IDR could be a transactivation domain (TAD).[8] We thus performed yeast two-hybrid assays in the Y187 strain in which protein–protein interactions were detected by the β-galactosidase activity of the *Lac Z* reporter gene. As expected, X-Gal activity showed that TCP8 formed dimers whatever the presence of the C-terminal IDR (Fig. 7). Some blue coloration was also observed with TCP8 alone in fusion with the GAL4 DNA binding domain (Fig. 7) while no X-Gal activity was detected with TCP8 alone in fusion with the GAL4 DNA activation domain (data not shown). TCP8 was thus able to activate the reporter gene by itself. This activation disappeared when *δC/TCP8* was fused to the GAL4-binding domain instead of full-length *TCP8* (Fig. 7). Therefore, the C-terminal part of TCP8 corresponds effectively to a TAD.

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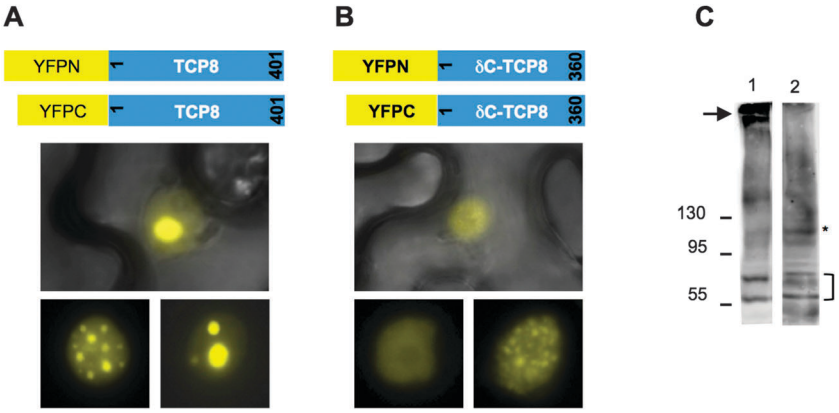

Fig. 6 Role of the C-terminal domain of TCP8 in aggregation of the dimer. *N. benthamiana* leaves were infiltrated with a mix of *Agrobacterium tumefaciens* transformed with BiFC plasmids carrying the full-length cDNA of *TCP8* (A) or the truncated form *δC/TCP8* (B) in frame with the N-terminal part or the C-terminal part of YFP. Top rectangular images, overviews of YFP fluorescence 3 d after infiltration and bottom panels, two representative views of YFP fluorescence observed by epifluorescence microscope. Views are representative of more than 3 biologically independent experiments. (C) Immunodetection of TCP8-YFP and δC/TCP8-YFP proteins in infiltrated *N. benthamiana* leaves. Proteins from infiltrated leaves with full length cDNA of *TCP8* (lane 1) and truncated cDNA of *TCP8* (lane 2) were extracted with a native buffer and separated by SDS-PAGE and subsequently probed with anti-TCP8. The arrow, the asterisk and the bracket correspond to aggregates, dimers and monomers of TCP8 or δC/TCP8, respectively.

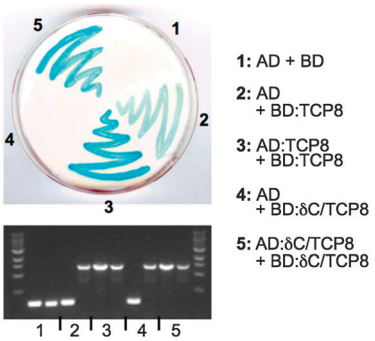

Fig. 7 The C-terminal IDR is a transactivation domain. The activation of β-galactosidase reporter gene gives a blue color in the presence of X-Gal in yeast two-hybrid assays. Numbers 1 to 5 refer to DNA constructs present in each transformed yeast strain. The presence or absence of the insert for each transformed yeast colony was confirmed by PCR, using specific primers for the activation domain (AD, left lane) and the binding domain (BD, right lane). Data shown are representative of 3 biological independent experiments.

Discussion

This study deals with the TCPs transcription factors family in *Arabidopsis thaliana* and particularly, focuses on TCP8, a class I TCP.[42] *In silico* analysis showed, for the first time, that all the TCPs members have a biased amino acid composition typical of IDPs. All TCPs, except TCP11, contain long IDRs (≥30 consecutive disordered residues) that are spread out over the sequence. These IDRs are not restricted to the N-terminal part as shown for the GRAS family, or to the C-terminal part as described for the NAC TF family and the CRY proteins family.[31,35,38] PONDR predictor software shows that there is more overall disorder in class I TCPs than in class II TCPs. While in *Arabidopsis thaliana* TFs, the overall disorder is about 46.64%,[8] and for class I and class II TCPs it is about 55.7% and 43.26%, respectively. In that context, TCP8 appears particularly interesting with an overall disorder of 63%. This prediction was confirmed by experimental data. The apparent molecular mass of TCP8 in SDS-PAGE (57 kDa) was higher than the theoretical one (47 kDa), a common feature of IDPs. The tendency of TCP8 to be degraded during purification might probably also be related to its intrinsic disordered nature since IDPs are extremely sensitive to proteolysis *in vitro*.[66] Moreover, hydrodynamic properties of TCP8 indicated that this protein corresponds to a molten globule-like IDP that is the most folded kind of IDPs. Disorder in TCP8 is mainly located in three regions of more than 50 residues. The percentage of TFs predicted to contain such long IDRs is 56% in *Arabidopsis thaliana*.[8]

The first IDR (54 residues) at the N-terminal part overlaps the TCP domain and particularly, the basic DNA binding domain (DBD). This DBD has been described as one of the most disordered domains among DBDs with AT hook domains.[8,37] Extensive studies have been performed on the DNA binding ability and specificity of TCPs[41,52,55–57,67–74] and folding upon binding to the major groove of DNA has been proposed for TCP4 DBD.[70] Recently, Hammani *et al.* have shown that TCP8 can bind site II elements in the promoter of *PNM1* (for Pentatricopeptide repeat protein localized to the Nucleus and Mitochondria 1) using electromobility shift assays.[49,75] Nonetheless, the binding mechanism was not investigated.

This N-terminal IDR starts with a region highly enriched in Thr and Ser residues. The percentage of Ser and Thr residues was 9.8% and 24.4%, respectively while frequencies of 6.3% and 6.15% were given for SwissProt proteins. This enrichment in phosphorylable residues is a common feature shared by the three IDRs of TCP8. Indeed, the abundance of Ser residues reached 8.6% and 14.9% in the IDRs present in the middle and the C-terminal parts of TCP8, respectively. Iakoucheva *et al.* have shown that protein phosphorylation predominantly occurs within IDRs[61] and effectively, DEPP predictor software shows that putative phosphorylated Ser residues are mainly

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clustered in these three IDRs as also shown for GRAS or Remorin proteins, two classes of plant specific IDPs.[34,35] This prediction was confirmed by experimental data. Nano liquid chromatography-tandem mass spectrometry identified Ser-211 that belongs to the IDR present in the middle of the sequence as phosphorylated. Phosphorylation and dephosphorylation are crucial events to modulate IDPs/IDRs functions. This post-translational modification can either promote or impede the interactions between IDPs and their different targets.[14,37,76,77] It also allows a fine regulation of interacting partners and contributes to their localization.[26,78,79]

A highly IDR of 73 residues, enriched in Ser, Gln and Asn residues, has been predicted at the C-terminal part of the sequence. Yeast two-hybrid assays showed for the first time that this IDR corresponds to a transactivation domain (TAD). This is in good agreement with the high content in acidic residues and the high percentage of disorder.[8,80] Self-activation has been previously reported for several rice[41] and *Arabidopsis* TCPs[56,70] but not for TCP8. This IDR is not only a TAD but also is of prime importance for TCP8 oligomerization. Indeed, full length TCP8 is able to self-assemble in dimers, trimers and multimers in a concentration-dependent manner while TCP8 lacking the 40 last residues can no longer form high order edifices as shown by BiFC experiments. Nonetheless, yeast two-hybrid assays and BiFC experiments showed that the truncated form still dimerized. As the HLH part of the TCP domain is described to be responsible for the dimerization of the TCP members, this result was not surprising.[42,43,70] Oligomerization has already been described for several IDPs[81,82] and the CC domain that was predicted inside this C-terminal IDR, could be responsible for oligomerization. Indeed, pronounced self-association *via* CCs has been largely described in the literature.[60,62,83–85] Nonetheless, multimerization observed with TCP8 is not only due to the CC present in the C-terminal IDR since speckles are still observed in BiFC experiments when it is removed. Thus, this domain tends rather to increase aggregation of TCP8 dimers. This behavior is similar to that observed with the N-terminal domain of CK2β1, the maize regulatory subunit of protein kinase CK2.[86] This limited aggregation could then be explained by the high proportion of Gln and Asn, described as aggregation-prone residues.[87] BiFC experiments have also shown that TCP8 aggregation is absent when it interacts with TCP15 that could stabilize the TCP8 monomer in an aggregation-incompetent conformation.

IDPs or IDRs are characterized by plasticity allowing them to interact with several partners with high specificity and low affinity. They frequently bind to their partners through short recognition elements embedded in a disordered environment[18,88] and in some cases undergo binding-induced folding.[19,89] *In silico* analysis of TCP8 has allowed the identification of several putative recognition elements indicating that TCP8 could interact with different proteins. This work has effectively shown that TCP8 is able to interact with TCP15 while recently, Hammani *et al.* have identified an interaction between TCP8 and PNM1, a Pentatricopeptide Repeat Protein in *Arabidopsis thaliana* using yeast two-hybrid experiments.[49] Since TCPs pertain to the IDP family, promiscuity is expected and it has been effectively shown for some of them. TCP11 is able to dimerize with different TCPs and interacts with circadian clock proteins.[56,57] TCP15 also interacts with either TCP members or with proteins involved in cytokinin responses or circadian clock.[50,56,57] This promiscuity is not exclusive to class I TCPs and is also observed for class II TCPs.[54,56,69] MoRFs predicted in TCP8 are thus in good agreement with the high tendency of IDPs, and particularly TCP members, to interact with several partners.

Experimental

Protein sequence analyses and disorder predictions

Deviations in the amino acid composition of TCPs were calculated using the average amino acid frequencies of the Swiss-Prot database as reference values. Hydrophobic cluster analysis (HCA) and disorder predictions were performed with the MeDor program (http://vazymolo.org) and PONDR® (http://www.pondr.com). The Coils algorithm (http://www.ch.embnet.org) was used to predict coiled coils. MoRFs and protein binding regions were predicted by using MoRFpred (http://biomine.ece.ualberta.ca/MORFpred/) and ANCHOR (http://anchor.enzim.hu/). Phosphorylation prediction was performed with DEPP predictor (http://www.pondr.com). Access to PONDR® was provided by Molecular Kinetics (6201 La Pas Trail – Ste 160, Indianapolis, IN 46268; 317-280-8737; E-mail: main@molecularkinetics.com). VL-XT is copyright©1999 by the WSU Research Foundation, all rights reserved. PONDR® is copyright©2004 by Molecular Kinetics, all rights reserved.

Expression and purification of TCP8

TCP8 was cloned using the Univector Plasmid-fusion System (UPS)[90] in which plasmid fusion between the Univector and the pHost plasmid *via* the loxP site creates a plasmid dimer. Full length *Arabidopsis* cDNA of *TCP8* in UPS donor vector pUNI51 and host vector pHB3-His6 and pHI100-His6, were from the Arabidopsis Biological Resource Center (ABRC).[91] For recombination reactions, 100 ng of each donor and acceptor vectors were mixed with 1 U of CRE recombinase (Novagen/Merck, Darmstadt, Germany) in 20 μL reaction volume and incubated at 37 °C for 45 min. The reaction mix was used to transform heat-shock competent DH5α cells that were plated on LB medium with 25 μg mL^{-1} kanamycin and 100 μg mL^{-1} ampicillin. Positive clones were sequenced.

For bacterial expression, BL21(DE3) competent cells were transformed and grown in LB medium with 100 μg mL^{-1} ampicillin and 25 μg mL^{-1} kanamycin at 37 °C until OD_{600} was 0.9 to 1. After cooling cultures on ice, TCP8 expression was induced with 1 mM isopropyl-β,D-thiogalactopyranoside (IPTG) at 30 °C for 45 min. Bacteria were collected by centrifugation and stored at −80 °C.

For expression in insect cells, recombinant baculovirus containing the TCP8 cDNA was produced by co-transfecting confluent *Spodoptera frugiperda* (Sf9) cells with 2 μg of pHI100-His6-TCP8 vector and 500 ng of linearized baculogold viral DNA

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following the manufacturer's instructions (BD, Le Pont de Claix, France). After 4 d at 27 °C, the supernatant was collected and used to amplify the virus for 7 d at 27 °C. The virus obtained was stored at 4 °C. To express TCP8, confluent High Five™ (Hi5) cells were infected with the virus and grown in serum-free medium at 27 °C. Aliquots were taken daily and analyzed on western blots to monitor the expression of recombinant TCP8. Hi5 cell culture medium 72 h after infection was centrifuged and the pellet was stored at −80 °C.

Pellets were resuspended in Ni-NTA buffer (50 mM Tris–HCl pH 8, 150 mM NaCl, 10 mM imidazole, 10 mM β-mercaptoethanol) with 1% Triton X-100, benzonase (1 : 10 000 (v/v), Sigma-Aldrich, St-Quentin Fallavier, France), 10 mM $MgCl_2$, 0.1 mM PMSF, 5 μg mL^{-1} leupeptin and a protease inhibitors cocktail (Sigma-Aldrich, St-Quentin Fallavier, France). Cells were broken at 1.7 kbar with a cell disrupter (Constant System Ltd, UK) and centrifuged at 10 000*g* for 30 min. The supernatant was applied to a Ni-NTA column (Qiagen, Courtabœuf, France) equilibrated in Ni-NTA buffer supplemented with 0.1% (v/v) Triton X-100. The column was first washed with 100 mM imidazole, then eluted with 300 mM imidazole. The fractions containing TCP8 were pooled, concentrated and dialyzed against the diethylaminoethyl (DEAE) buffer (50 mM Tris–HCl, pH 8, 10 mM β-mercaptoethanol, 0.1% (v/v) Triton X-100). Proteins were then applied to a DEAE Trisacryl column (GE Healthcare, Vélizy, France) equilibrated in DEAE buffer. The column was washed with the DEAE buffer and eluted with a 0 to 0.5 M NaCl linear gradient. The purity and integrity of TCP8 were assessed by sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS-PAGE) and matrix-assisted laser desorption mass spectrometry (MALDI-TOF MS) at the IFR83 mass spectrometry core facility.

Recombinant TCP8 cross-linking

Purified recombinant TCP8 (0.03 mg mL^{-1}) was incubated at 37 °C for 6 min with 0.05% glutaraldehyde. The reaction was stopped by adding 0.1 M ethanolamine and samples were run on 6% SDS-PAGE gels.

Electrophoresis and western blot

Protein samples were run on SDS-PAGE or NATIVE-PAGE (Mini-PROTEAN Electrophoresis System, Bio-Rad, Marnes-la-Coquette, France). Proteins were electro-transferred onto nitrocellulose membrane for 1 h at 120 V (Mini Trans-Blot Cell, Bio-Rad, Marnes-la-Coquette, France). Membranes were incubated with 5% (w/v) powdered milk in Tris–Tween buffer saline (TTBS; 20 mM Tris–HCl pH 7.6, 150 mM NaCl, 0.1% (v/v) Tween 20) at room temperature for 1 h with shaking, then with purified polyclonal TCP8 antibodies (1 : 2000) or monoclonal hemaglutinin antibodies (HA; 1 : 1000, Eurogentec, Angers, France) in TTBS overnight at 4 °C with shaking. Affinity-purified antibodies raised against purified denatured TCP8, expressed in *E. coli*, were obtained from rabbits (GeneCust Europe, Dudelange, Luxembourg). Membranes were then washed and incubated with 1 : 8000 horseradish peroxidase (HRP) conjugated secondary antibodies, and developed by enhanced chemiluminescence (GE Healthcare, Vélizy, France).

Gel filtration and dynamic light scattering (DLS)

Filtrated and purified TCP8 was loaded on a Superdex 200 10/300 GL column (GE Healthcare) pre-equilibrated in Tris 50 mM, NaCl 0.15 M, EDTA 0.1 mM, DTT 1 mM pH 7.9. Proteins were eluted at a flow rate of 0.3 ml min^{-1}. Fractions of 0.5 ml were collected and measured for presence of proteins at 280 nm.

Four natively folded proteins (β-amylase (200 kDa), alcohol dehydrogenase (150 kDa), ovalbumin (43 kDa) and carbonic anhydrase (29 kDa)) were used as standards to calibrate the Superdex 200 10/300 GL column and their theoretical Stokes radii R_s^{NF} were calculated from the equation $\log(R_s^{NF}) = 0.369\log(M_r^{Theo}) - 0.254$, where M_r^{Theo} represents theoretical molecular weight. The experimentally determined migration rate ($1000/V_{elution}$) of each standard protein was plotted against theoretical Stokes radius and theoretical molecular weight. These plots were used to determine the experimental Stokes radius (R_s^{TCP8}) and apparent molecular weight (M_r^{App}) of TCP8. The theoretical Stokes radii of TCP8 in various folding states were derived from their theoretical molecular weight using the appropriate equations.[1]

In solution behavior of TCP8 was analyzed by DLS on a DynaPro instrument (Wyatt Technology Corp., Santa Barbara, CA). The laser light wavelength was 830 nm and the scattered light was collected at 90° by an optic fibre until a detector. Purified TCP8 (2 mg ml^{-1}) was filtered through 0.22 μm filters. The correlation time τ was 0.5 μs and the acquisition time was 10 s for a total measurement of 30 min. The sample temperature was fixed at 20 °C with a Peltier cell. The polydispersity (% P) was obtained using regularization methods (Dynamics software, version 6).

BiFC assay

TCP8 and *TCP15* cDNAs were amplified by PCR and the truncated version (δC/TCP8) of TCP8 was obtained by site directed mutagenesis according to the manufacturer's instructions (Life Technologies SAS, Saint Aubin, France). The specific primers used are detailed in Table S1 (ESI†).

Fragments were cloned in pSPYNE173, pSPYCE(M), pSPYNE(R)173 and pSPYCE(MR) kindly provided by Pr J. Kudla. pSPYNE173 and pSPYNE(R)173 encoded Myc epitope tag whereas pSPYCE(M) and pSPYCE(MR) encoded HA epitope tag.[92] All constructs were sequenced. *Agrobacterium tumefaciens* strain GV3103 (pMP90), kindly provided by Pr C. Koncz, was transformed with each plasmid by electroporation of 40 μL of electrocompetent cells with 40 ng of the vector (MicroPulser™ electroporator, Bio-Rad, Marnes-la-Coquette, France) then plated on LB-agar with 50 μg mL^{-1} kanamycin and 25 μg mL^{-1} gentamycin and grown at 28 °C. Transformed *A. tumefaciens* was grown in 50 mL LB medium with 50 μg mL^{-1} kanamycin, 25 μg mL^{-1} gentamycin, 10 mM MES–KOH pH 5.7 and 20 μM acetosyringone at 28 °C. Cells were collected by centrifugation (5000*g*, 20 min), resuspended in 10 mM MES–KOH pH 5.7, 10 mM $MgCl_2$ and 150 μM of acetosyringone. After 3 h at room

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temperature, the OD_{600} was measured and cultures diluted to OD_{600} 0.8 (following the protocol from the Pr D. Baulcombe laboratory web site, http://www.plantsci.cam.ac.uk/research/davidbaulcombe.html).

Three-week-old *N. benthamiana* plants were infiltrated with a mix of four different *A. tumefaciens*, two carrying the plasmid with the N- or C-terminal part of YFP in frame with the corresponding TCP cDNA, the third carrying the pBIN61-p19 plasmid (kindly provided by Pr D. Baulcombe) to suppress RNA interference[93] and the fourth carrying the pROK2-*AtFib2-mRFP* to visualize the nucleolus.

After infiltration, the abaxial side of leaves was observed by epifluorescence microscopy (Imager Z1, Zeiss,) using laser excitation at 505 nm for YFP and 555 nm for mRFP.

Protein extraction

Agroinfiltrated leaves (100 to 400 mg) were ground to a powder in liquid nitrogen with 4.8 mm stainless steel beads in a mixer mill (MM301, Retsch, Haan, Germany). One volume of extraction buffer (100 mM Tris–HCl pH 8, 4% (v/v) SDS, 20% (v/v) glycerol) was added. Extracts were shaken thrice for 10 min each in a multi-head agitator at 1000 rpm with 10 min intervals on ice. The extract was centrifuged 20 min at 8000*g* and the protein concentration of the supernatant was measured with the Folin–Lowry method.[94]

To study TCP8 aggregation in agroinfiltrated leaves, the same protocol was used except that the extraction buffer was made of 100 mM Tris–HCl pH 8, 0.5 mM EDTA, 0.5 mM EGTA, 0.5 M NaCl, 0.5% (v/v) Triton X-100, 10 mM DTT, 0.1 mM PMSF, 5 μg mL^{-1} leupeptin and a protease inhibitor cocktail (Sigma-Aldrich, Saint-Quentin Fallavier, France). Before centrifugation, cells were broken at 1.5 kbar with a cell disrupter.

Yeast two-hybrid assay

Full-length or truncated cDNA of *TCP8* were amplified by PCR using the primers described in Table S1 (ESI†). Amplified products were cloned in pGBT9 (bait) and pGAD424 (prey) plasmids (Clontech, Takara Bio, Lonza BioScience, Saint-Beauzire, France). Y187 (Matα, ura3-52, his3-200, ade2-101, trp1-901, leu2-3, 112, gal4Δ, gal80Δ, met-, URA3::$GAL1_{uas}$-$GAL1_{TATA}$-lacZ) yeast strain was co-transformed following the TRAFO protocol (http://home.cc.umanitoba.ca/~gietz/2HS.html). Overnight cultures (1.5 mL) of Y187 in yeast peptone dextrose adenine (YPDA) medium were centrifuged for 1 min at 850*g*. Pellets were rinsed with 1 mL of H_2O and resuspended in 200 μL of 50% polyethylene glycol-4000 solution (PEG-4000 in 10 mM Tris–HCl, pH 8.0, 1 mM EDTA (TE) and 0.1 M LiOAc), 20 μL of DMSO and 20 μL of H_2O. The mix was vortexed until clumps dispersed then 1.2 μg of plasmid and 10 μL of salmon sperm (10 mg mL^{-1}) were added. The mix was shaken for 20 min at 30 °C and heat-shocked for 15 min at 42 °C. Sterile water (1 mL) was added and the suspension was centrifuged for 1 min at 850*g*. The pellet was resuspended in 1 mL of water and 200 μL was plated on 2% glucose synthetic defined (SD) medium containing the amino acid residue dropout (DO) without leucine (-Leu) or tryptophan (-Trp). For the second transformation, overnight cultures of these colonies were diluted 1/10 and incubated 4 h at 30 °C. Four mL of this culture was transformed following the above procedure. Co-transformed yeasts were plated on SD with DO-Leu-Trp.

X-Gal assay

Colonies were grown for 2 d on a nitrocellulose membrane (Hybond, GE Healthcare, Vélizy, France) then placed on SD with DO without leucine (-Leu) and tryptophan (-Trp). Cells were broken by placing the nitrocellulose in liquid nitrogen for 10 s then on Whatman paper imbibed with the 'buffer Z/X-Gal solution' as described in Yeast Protocols Handbook (Clontech, 2009). Membranes were scanned every hour to check for the development of the blue color.

Mass spectrometry analysis of TCP8 expressed in insect cells

In-gel proteins were treated with steps of washing, reduction–alkylation of cysteine residues and digestion by trypsin as previously described.[95] For MALDI-TOF-TOF mass spectra analysis using an Applied Biosystems 4700 Analyzer (Applied Biosystems, Courtabœuf, France), samples were desalted on C_{18} Zip tips (Millipore, Molsheim, France) and eluted in 1 μL α-cyano-4-hydroxycinnamic acid as matrix. For the identification of phosphorylated peptides, IMAC enrichment of phosphopeptides was performed with the Clontech spin column kit (Ozyme, France) according to the manufacturer's protocol using two elution steps. Mass spectrometry was performed on a LTQ Velos Orbitrap mass spectrometer (Thermo Fisher, USA) equipped with a nanospray ion source and coupled to nano-flow high performance chromatography (Ultimate NCS3500, Dionex, France). Tryptic peptides were dissolved in the loading buffer (2% acetonitrile, 0.05% TFA in water) and desalted on a C18 nano-trap (Acclaim PepMap100, 100 μm × 2 cm, 5 μm, 100 Å, Dionex) mounted in a 6-port valve before elution onto a C18 column (Acclaim PepMap RSLC, 75 μm × 150 mm, 2 μm, 100 Å, Dionex). Peptides were eluted with a linear gradient from 4% to 55% of mobile phase B (20% water, 80% acetonitrile, 0.1% formic acid) in A (0.1% formic acid in water) for 30 min. The peptides were detected in the mass spectrometer in positive ion mode using a full MS at R = 30 000 in the Orbitrap as the first scan event, followed by high collision dissociation (Energy 45%) using Data Dependent MS/MS mode. Spectra were processed with Proteome Discoverer software (Thermo Fisher Scientific) and protein searches were performed with MASCOT (*in situ* license) and OMSSA (MSDA, LSMBO Strasbourg facility). For the Mascot search, the following parameters were used: SwissProt 2011 reduced to *Arabidopsis thaliana* (10 617 sequences); enzyme – trypsin; variable modifications – carbamidomethylation (C), oxidation (M), phosphorylation (Ser, Thr, Tyr); mass values – monoisotopic; protein mass – unrestricted peptide; mass tolerance – ±10 ppm; fragment mass tolerance – ±0.8 Da; missed cleavages. For the OMSSA search, the SwissProt database reduced to *Spodoptera frugiperda* including TCP8 sequence was used with the same parameters as described above, except that MS/MS tolerance was reduced to ±0.3 *m/z*. In all searches, proteins were considered

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as identified by 2 unique peptides with ion significance threshold $p < 0.05$ and protein score higher than 30.

Conclusion

In conclusion, the work herein clearly supports the proposal that the plant-specific TCP family of transcription factors pertain to the IDP family. In particular, this work on *A. thaliana* TCP8 contributes to explain the presence of TCP in so many physiological processes. As disorder is intimately linked to promiscuity, it allows the interaction of TCPs with a myriad of partners and consequently a fine regulation of the transcription in response to different stimuli. As study of the TCP family is an emerging topic, there is no doubt that more interactions will be discovered in the future.

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